



Speaker-individuality in suprasegmental temporal features: Implications for forensic voice comparison



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ABSTRACT

Everyday experience tells us that it is often possible to identify a familiar speaker solely by his/her voice. Such observations reveal that speakers carry individual features in their voices. The present study examines how suprasegmental temporal features contribute to speaker-individuality. Based on data of a homogeneous group of Zurich German speakers, we conducted an experiment that included speaking style variability (spontaneous vs. read speech) and channel variability (high-quality vs. mobile phone-transmitted speech), both of which are characteristic of forensic casework. Speakers demonstrated high between-speaker variability in both read and spontaneous speech, and low within-speaker variability across the two speaking styles. Results further revealed that distortions of the type introduced by mobile telephony had little effect on suprasegmental temporal characteristics. Given this evidence of speaker-individuality, we discuss suprasegmental temporal features' potential for forensic voice comparison.

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1. Introduction

When participating in a conversation with a group of people, we often have no trouble segregating speakers by their voices even if we have never met these speakers before. Experience also shows that we can typically identify an acquaintance on the telephone after only a few syllables. These examples illustrate that human voices are highly individual. The phenomenon that the speech signal contains speaker-individual information is exploited in speaker identification and verification procedures [1] and in particular in forensic voice comparison (hereafter FVC; [2]). In typical FVC cases, acoustic trace material from a crime (normally recordings of a perpetrator) are compared to acoustic comparison material (typically recordings of a suspect), and used for post-crime forensic investigations.

Voices can be individual in different acoustic domains. Research on speaker-individuality tended to focus on the frequency domain (fundamental frequency: [3–7]; formant frequencies: [8–16]) and the intensity domain of speech [17]. Relatively little attention has been paid to speaker-specific temporal characteristics. The principal objective of the present study is to examine in further

detail the speaker-individuality of temporal features. Here we focus in particular on suprasegmental temporal features, that means temporal features of speech that are not restricted to a single segment (for example a consonant or a vowel, cf. [18]) but to the more global temporal organization of speech in an utterance. Such temporal organization of speech has traditionally been referred to as *speech rhythm*. For this reason, the aim of the present approach was to use measures that are frequently used in the field of speech rhythm.

Why do we focus on suprasegmental temporal characteristics?

- (1) Effects of between-speaker suprasegmental temporal variability were observed for a number of different datasets in the field of speech rhythm [19–23]. These results, however, are based on datasets that were designed to investigate between-language effects [23] or they are based on a small number of speakers and on speech material in which possible between-speaker artifacts such as accent or dialect were not carefully controlled for [19–22]. In the present study, we investigated speaker-individual suprasegmental temporal features for 16 speakers that were highly controlled for accent (Zurich German), age (20–30 years), and social background (university students).
- (2) Recent evidence points to two possible sources for speaker-individuality in suprasegmental temporal features: speaker idiolect and speaker anatomy. Speakers vary in an acquired way they use speech, having their own way of lengthening

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sound patterns or having a preference towards certain syllables or sound segments. In terms of speaker anatomy, albeit coming from different strains of research, research showed that human movement is highly individual ([24,25] for gait; [26,27] for typing-movements). Eriksson and Wretling [28] suggested a comparable stability of timing patterns in human speech, which is in the same way produced by intricate, brain-controlled muscle movements as leg or finger movements found in walking and typing. It thus seems plausible that similar individualities as in human gait or finger movements are also present in articulatory movements.

In typical forensic phonetic casework the phonetic expert compares a number of speaker-individual characteristics between acoustic trace and comparison material. Such characteristics can either be on born features of the vocal tract such as voice fundamental frequency and vocal tract resonance characteristics or acquired features such as accent, dialect or sociolectal ways of pronunciation. It is essential for FVC to accumulate as many characteristics of the speech signal as possible [1,2]. With our research we aim at examining further speaker-individual information that may be used in FVC in the future. For such an application it is essential to have in-depth knowledge about the variability of the characteristics under scrutiny within and between speakers and about how such variables are affected by different signal transmission conditions (e.g. mobile phone). It is desirable for forensic circumstances that the features under investigation reveal maximum between-speaker variability and minimum within-speaker variability [1].

In forensic phonetic casework, trace and suspect material is either spontaneously produced or read. At the German Federal Criminal Police Office (BKA), an estimated 10–20% of trace material is read, while a somewhat larger amount, 10–30%, of suspect material is read (Olaf Köster, BKA, personal communication). The vast majority in both trace and suspect material, however, is spontaneously produced. Spontaneous and read speech differ on various levels: the former is optimized for human-to-human communication, shows simultaneous planning and execution, and overall demonstrates greater segmental and suprasegmental variability [19,29–31]. Since trace and suspect material often differ in speaking style, it is critical to know whether such differences affect the speech parameter being evaluated in FVC.

Aside from possible differences in speaking styles, trace and suspect material may also differ in channel transmission. 90% of the time, forensic trace and suspect material involves telephone-transmitted speech [32]. Telephone-transmitted speech is different from high-quality recorded speech in that it features band-pass transmission channels of only 350–3400 Hz [33,34], higher F1s, more narrow dynamic ranges, and artefactual peaks [33–36]. Mobile phone-transmitted speech shows wider variability in the transmission quality and more restrictive band pass filters, causing F1 in close and mid vowels to be even higher than over landline-telephone-transmitted speech [33]. Such technical effects compromise the reliability of frequency-based measures. It is likely that suprasegmental temporal features are advantageous in this respect: the points in the speech signal where vowels or consonants start, or where voicing starts or ends, should largely remain unaffected by the technical effects of mobile phone transmission. In this regard, suprasegmental temporal measures may be able to enhance FVC analyses particularly when the speech signal is degraded.

In a within-subject design, we examined the above-mentioned types of variability pertinent to forensic phonetics: speaking style variability (spontaneous vs. read) and channel variability (high-quality vs. mobile phone-transmitted). The design contained the same linguistic material, i.e. sentences, for each speaker and condition since temporal characteristics are known to be sensitive

to sentence material [20,23]. The present study addresses the following research questions:

1. Are there between-speaker differences in suprasegmental temporal features? (see Section 3.1).
2. Which suprasegmental temporal measure explains most variation between speakers? (see Section 3.2).
3. How robust are suprasegmental temporal features to speaking style variability? (see Section 3.3).
4. How robust are suprasegmental temporal features to channel variability? (see Section 3.4).

Given the discussion above we expect to find significant between-speaker variability in both read and spontaneous speech as well as little within-speaker variability across the two speaking styles. Moreover, channel variability is likely to have little effect on suprasegmental temporal features.

2. Methods

2.1. Speakers

16 speakers (8 male/8 female) of Zurich Swiss German were recorded in a sound-treated booth at the Phonetics Laboratory of the University of Zurich. Eligibility criteria required individuals to demonstrate little to no regional and social accent variability. Average age was 27, SD = 3.6, and age range 20–33. None of the speakers reported hearing or speech disorders. The data was recorded in a sound-treated booth using a Neumann STH-100 transducer microphone (sampling rate of 44.1 kHz; 16 bit quantization).

2.2. Material

2.2.1. High-quality spontaneous speech

In a first recording session, spontaneous speech material was collected via semi-structured interviews. The 16 speakers were asked to talk freely to the interviewer (first and second author) about their studies at the University of Zurich. The interview was conducted in Swiss German. A subset of sentences, 16 per speaker (typically 15–20 syllables per sentence), was isolated from these interviews. For the isolation of sentences there are no formal criteria that allow for an identification of utterances as complete “units”. We selected sentences according to syntactic, prosodic, voice quality, pausing and breathing criteria [37]. The isolated sentences had to form meaningful units and be fluently spoken, i.e. free from filled and unfilled pauses, hesitations, and mispronunciations. These 256 sentences (16 speakers × 16 sentences) formed the spontaneous, high-quality (henceforth *hifi*), corpus of this study.

2.2.2. High-quality read speech

We made orthographic transcripts of these 256 spontaneous sentences. These transcripts were given to the same 16 speakers with the request to prepare reading the sentences for a second recording session. Approximately three months after the first session, the 16 speakers read those 256 sentences in our laboratory (16 previously self-produced sentences + 240 sentences from their peers). It is plausible to assume that, given the temporal discrepancy between the two recording sessions, the obtained effects may either be attributed to speaking style variability or to the temporal delay between the two sessions. There is evidence from previous research, however, showing no effect of *test-retest* for suprasegmental temporal features [38]. These 4096 sentences (256 sentences × 16 speakers) constituted the read, *hifi* corpus of this study. The spontaneous and read *hifi* corpora amount to 56,794 syllables in total.

2.2.3. Mobile phone-transmitted spontaneous speech

For all 16 speakers, the 16 spontaneous sentences were transmitted via a 3G GSM phone line. The sentences were played on a computer with external speakers (AppleDesign Powered Speakers; M6082). A mobile telephone device (Apple iPhone 4S) was placed 1.5 cm from the center of one of the speakers. In an adjacent closed-off room, a second mobile telephone device (Apple iPhone 5) received the call. The amplitude of the output signal was adjusted so that the signal was maximally loud without clipping. The signal was directly fed into the line-in of a digital recording device (ZOOM H2) via 3.5 mm audio jack. A mono recording was carried out which was sampled at 44,100 samples/second a quantization depth of 16 bit.

2.2.4. Data editing

Trained phoneticians (first and second author) attributed the segmental on- and offset information to the corpus in Praat [39] and transcribed each segment using SAMPA phonetic alphabet. The labelers cross-checked their segmentation in regular intervals so as to maximize inter-labeler consistency. Based on this phone labeling, automatic processing of consonantal and vocalic intervals was carried out. Syllables were automatically annotated according to sonority hierarchy principles [40]. Sonority decreases from the syllable nuclei to the syllable edges. Following this decrease, sonority increases again, which is the locus where syllable boundaries were placed. Unlike in German, syllabification in Swiss German dialects does not strictly proceed according to morpheme boundaries [41,42]. Swiss German follows the onset-maximization principle: syllable onsets are maximized before syllabification proceeds to the coda and nucleus of the preceding syllable [43]. This automatic syllabification was adjusted manually.

2.3. Temporal measures applied

We used a wide variety of temporal measures that are commonly used in the field of speech rhythm research: we measured durational variability of consonantal and vocalic intervals, voiced and unvoiced intervals, and syllable-peak-to-syllable-peak intervals. The use of syllable peaks is grounded in P-center theory [44–46], which claims that perceptually prominent centers of a syllable exist in the vicinity of vowel intensity peaks in the syllable nucleus. This must be viewed as an approximation of the P-center. Alternative models show that vowel onsets, for example, additionally contribute to the perception of rhythmic beats [47]. For the present model, however, we worked with a measure based on amplitude peaks only. Overviews of most of these measures are given in ref. [48], further mention of measures (9) and (10) (below) in ref. [49].

Five measures based on vocalic and consonantal intervals

- (1) The percentage over which speech is vocalic (%V [50]).
- (2) The rate-normalized standard deviation of vocalic interval durations (VarcoV [51])

$$\text{VarcoV} = 100 \times \frac{\Delta V}{\bar{V}}$$

where ΔV is the standard deviation of vocalic interval durations and $\bar{V}$ the mean vocalic interval duration.

- (3) The rate-normalized average differences between consecutive vocalic interval durations (nPVI_V [52])

$$\text{nPVI}_V = 100 \times \left[\frac{\sum_{k=1}^{m-1} |(d_k - d_{k+1}) / ((d_k - d_{k+1}) / 2)|}{m - 1} \right]$$

where m is the number of vocalic intervals and d_k the duration of the k th interval.

- (4) The rate-normalized standard deviation of consonantal interval durations (VarcoC [53])

$$\text{VarcoC} = 100 \times \frac{\Delta C}{\bar{C}}$$

where ΔC is the standard deviation of consonantal interval durations and $\bar{C}$ the mean consonantal interval duration.

- (5) The rate-normalized average differences between consecutive consonantal interval durations (nPVI_C [52])

$$\text{nPVI}_C = 100 \times \left[\frac{\sum_{k=1}^{m-1} |(d_k - d_{k+1}) / ((d_k - d_{k+1}) / 2)|}{m - 1} \right]$$

where m is the number of consonantal intervals and d_k the duration of the k th interval.

Three measures based on voiced and voiceless intervals

- (6) The percentage over which speech is voiced (%VO [54]).
- (7) The rate-normalized standard deviation of voiced interval durations (VarcoVO [54]) The rate-normalized standard deviation of voiced interval durations (VarcoVO [54])

$$\text{VarcoVO} = 100 \times \frac{\Delta VO}{\bar{VO}}$$

where ΔVO is the standard deviation of voiced interval durations and $\bar{VO}$ the mean voiced interval duration.

- (8) The rate-normalized average differences between consecutive voiced interval durations (nPVI_VO [54])

$$\text{nPVI}_VO = 100 \times \left[\frac{\sum_{k=1}^{m-1} |(d_k - d_{k+1}) / ((d_k - d_{k+1}) / 2)|}{m - 1} \right]$$

where m is the number of voiced intervals and d_k the duration of the k th interval.

Two measures based on intervals between syllable peaks

- (9) The rate-normalized standard deviation of syllable-peak-to-syllable-peak interval durations (VarcoPeak [49])

$$\text{VarcoPeak} = 100 \times \frac{\Delta(\text{peak-to-peak})}{\bar{\text{peak-to-peak}}}$$

where $\Delta(\text{peak-to-peak})$ is the standard deviation of syllable-peak-to-syllable-peak interval durations and $\bar{\text{peak-to-peak}}$ the mean syllable-peak-to-syllable-peak interval duration.

- (10) the rate-normalized average differences between consecutive syllable-peak-to-syllable-peak interval durations (nPVI_Peak [49])

$$\text{nPVI}_\text{Peak} = 100 \times \left[\frac{\sum_{k=1}^{m-1} |(d_k - d_{k+1}) / ((d_k - d_{k+1}) / 2)|}{m - 1} \right]$$

where m is the number of syllable-peak-to-syllable-peak intervals and d_k the duration of the k th interval.

Voicing detection for the measures (6), (7), and (8) was performed automatically with the pitch detection algorithm in Praat [39] (standard settings). The algorithm for (9) and (10) captures syllable-intensity-peak-to-syllable-intensity-peak durations based on automatically annotated syllable intervals (cf. Section 2.2). The temporal measures were calculated sentence-by-sentence using *durationAnalyzer.praat*, a Praat script available under http://www.pholab.uzh.ch/static/volker/software/plugin_durationAnalyzer.zip.

2.4. Statistical analysis

All data were analyzed using R [55] and the R packages *lme4* [56] and *languageR* [57,58], as well as JMP [59]. If not indicated otherwise, we analyzed data using linear mixed effect models (LMEs). Normality was checked by visual inspection of quantile plots. *Speaker* and *sentence* were treated as random effects, *speaking style* and *channel* as fixed effects. Effects were tested by model comparison between a full model in which the factor in question is entered as either a fixed or a random effect (R code example: `model_full = lmer(dependent_variable ~ fixed_factor + (1|random_factor1) + (1|random_factor2), data = data)`) and a reduced model in which the factor in question is excluded (R code example: `model_reduced = lmer(dependent_variable ~ 1 + (1|random_factor1) + (1|random_factor2), data = data)`). *p*-Values were obtained by comparing the results from the two models using standard ANOVAs (R code: `anova(model_full, model_reduced)`). For the assessment of the relative goodness of fit we indicate AIC (Akaike Information Criterion) values, which decrease with goodness of fit [60]. Only *p*-values that are considered significant at the $\alpha = 0.05$ level are reported.

3. Results

The results section addresses each of the research questions posed in Section 1. Note that not every measure mentioned in Section 2.3 is applied in all analyses. For research question 3 *How robust are suprasegmental temporal features to speaking style variability?* only %V (1) and %VO (6) are calculated. As will be shown below, these two measures revealed most differentiation between speakers (cf. Section 3.2). For research question 4 *How robust are suprasegmental temporal features to channel variability?* we only calculated %VO (6) and VarcoPeak (9). Both measures are automatic measures yet they are conceptually different. Given this difference, it will be interesting to see how these two measures perform on mobile phone-transmitted speech. A further note on the factor *gender*. Based on read speech material (256 sentences per speaker), we performed LMEs for each temporal measure, comparing models with *gender* as fixed effect and *speaker* and *sentence* as random effects to their reduced models with no fixed effect and *speaker* and *sentence* as random effects. Results showed no effect of *gender*. Only the models for VarcoVO differed significantly from one another, with the model that includes *gender* providing a better fit ($AIC = -3307$, $p = .02$). Females exhibited less voicing variability ($M = .67$, $SD = .19$) than men ($M = .70$, $SD = .21$). Results were thus generalizable for the factor *gender*—which is why *gender* was not considered in the LMEs to follow.

3.1. Are there between-speaker differences in suprasegmental temporal features?

First, findings on read speech are reported (the same 256 sentences for each speaker), followed by findings on spontaneous speech (16 sentences taken from each speaker's own interview, therefore differing between speakers).

3.1.1. Read speech

Table 1 presents the results obtained from the model comparisons on the ten tested temporal measures.

All comparisons between full and reduced models showed a significant difference and all full models exhibited an increased goodness of fit, i.e. between-speaker variation was significant. This improvement occurred on every temporal level: vocalic and consonantal intervals, voiced and voiceless intervals, and syllable-peak-to-syllable-peak intervals.

Table 1

Summary of the statistics for the tested suprasegmental temporal measures for read speech.

Temporal measure	Test	Factor tested	Result
%V	LME	Speaker	$p < 0.0001$, AIC = 20196
VarcoV	LME	Speaker	$p < 0.0001$, AIC = -7482
nPVI_V	LME	Speaker	$p < 0.0001$, AIC = 30863
VarcoC	LME	Speaker	$p < 0.0001$, AIC = -9150
nPVI_C	LME	Speaker	$p < 0.0001$, AIC = 29969
%VO	LME	Speaker	$p < 0.0001$, AIC = 23980
VarcoVO	LME	Speaker	$p < 0.0001$, AIC = -3304
nPVI_VO	LME	Speaker	$p < 0.0001$, AIC = 35105
VarcoPeak	LME	Speaker	$p < 0.0001$, AIC = -7388
nPVI_Peak	LME	Speaker	$p < 0.0001$, AIC = 34155

3.1.2. Spontaneous speech

For each temporal measure we calculated a univariate ANOVA *speaker* by temporal measure. We ran ANOVAs instead of LMEs because LMEs cannot be calculated if the number of observations per speaker (16) is equal or less than the number of speakers (16). Table 2 shows the summary of these statistics.

Only %V and %VO showed significant effects of *speaker*, with the full models exhibiting an increased goodness of fit.

3.2. Which suprasegmental temporal measure explains most variation between speakers?

First, findings on the read speech are reported, followed by findings on spontaneous speech (same material as in Section 3.1).

3.2.1. Read speech

To assess which temporal measure explains most variation in read speech, we constructed a nominal logistic regression model. By inspecting likelihood-ratio tests in the model output we can evaluate the relative contribution of each predictor, i.e. each temporal measure, towards explaining temporal variability between speakers. The likelihood-ratio chi-square value of each variable provides the additional variability explained when this effect is added to the model [61]. The relative importance of an effect in a logistic regression is thus defined as the likelihood-ratio chi-square value of the effect, divided by the sum of the likelihood-ratio chi-square values of all effects, times one hundred. The regression was significant ($R^2 = .079$, $X^2 = 1793$, $p < .0001$, with $df = 150$). Note that in categorical data analysis, high R^2 values are rare [61]. Moreover, maximizing R^2 values is not of principal interest in this study. We are primarily interested in the individual contribution of each measure towards explaining variation in suprasegmental temporal features. The relative importance of the predictors in the model is shown in Fig. 1.

Table 2

Summary of the statistics for the tested suprasegmental temporal measures for spontaneous speech.

Temporal measure	Test	Factor tested	Result
%V	One-way ANOVA	Speaker	$F(15, 255) = 3.2$, $p < .0001$
VarcoV	One-way ANOVA	Speaker	ns.
nPVI_V	One-way ANOVA	Speaker	ns.
VarcoC	One-way ANOVA	Speaker	ns.
nPVI_C	One-way ANOVA	Speaker	ns.
%VO	One-way ANOVA	Speaker	$F(15, 255) = 2.3$, $p = .004$
VarcoVO	One-way ANOVA	Speaker	ns.
nPVI_VO	One-way ANOVA	Speaker	ns.
VarcoPeak	One-way ANOVA	Speaker	ns.
nPVI_Peak	One-way ANOVA	Speaker	ns.

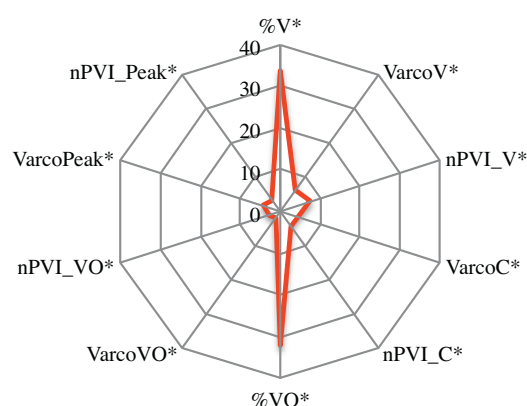


Fig. 1. Radar chart illustrating the individual contribution of the ten predictors in the nominal logistic regression model for *speaker* based on read speech.

The radar chart shows the breakdown of each predictor towards explaining suprasegmental temporal variability between speakers. Each radius represents one temporal measure. Measures with an asterisk generated significant effects in the regression model. The length of the radius is proportional to the magnitude of the factor in the created model, where nPVI_V, for example, explains 7% of the temporal variability. The strongest effects were found for the measures %V ($X^2 = 603$, $p < .0001$) and %VO ($X^2 = 576$, $p < .0001$), which explained 33% and 32% of the temporal variability between speakers.

%V and %VO are conceptually related: %V is the percentage over which speech is vocalic and %VO is the percentage over which speech is voiced. Vocalic segments are normally voiced, non-vocalic segments can be voiced or voiceless. In Zurich German, the vast majority of consonants are voiceless [42], except for nasals, trills and approximants. We processed a correlation of %V and %VO. The two variables were correlated ($r(4096) = .62$, $p < .0001$; $R^2 = .39$, $F(1, 4094) = 2609$, $p < .001$): %V explains 39% of the variability in %VO.

3.2.2. Spontaneous speech

We further constructed a nominal logistic regression model on the spontaneous speech data. The individual weight of the ten predictors in the regression is illustrated in the radar chart in Fig. 2. The regression was significant ($R^2 = .18$, $X^2 = 260$, $p < .0001$, with $df = 150$). The R^2 value (.18) was higher than the one obtained from the read data. The strongest effects were again found for %V

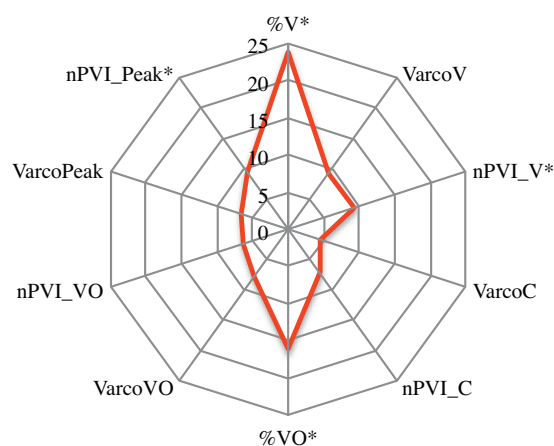


Fig. 2. Radar chart illustrating the individual contribution of the ten predictors in the nominal logistic regression model for *speaker* based on spontaneous speech.

($X^2 = 65$, $p < .0001$) and %VO ($X^2 = 44$, $p = .0001$), which explained 24% and 16% of the temporal variability between speakers. nPVI_Peak and nPVI_V both accounted for 9% (nPVI_Peak: $X^2 = 25$, $p = .043$; nPVI_V: $X^2 = 25$, $p = .046$).

3.3. How robust are suprasegmental temporal features to speaking style variability?

We report findings of 32 sentences per speaker (16 spontaneous sentences: taken from each speaker's own interview, therefore differing between speakers; 16 read sentences: the read counterparts per speaker, analogously differing between speakers). As for %V, the model with *speaker* as random effect provided an improved goodness of fit ($AIC = 3123$, $p = .003$) which means that between-speaker variation was significant. There was no significant difference in %V between the two speaking styles ($p = .76$) but there was a significant interaction of *speaker* $\times$ *style* ($AIC = 3090$, $p < .0001$). Fig. 3 shows the boxplots of the 16 speakers' %V for the spontaneous sentences ('spnt', yellow) and for the identical read counterparts ('read', blue). Non-overlapping notches in the boxes suggest that the medians differ.

To test for the simple effect of *speaker* we processed two ANOVAs: one on the spontaneous sentences and one on the read sentences. We ran ANOVAs instead of LMEs because LMEs cannot be calculated if the number of observations per speaker (16) is equal or less than the number of speakers (16). Both ANOVAs showed significant effects of *speaker* (Bonferroni adjusted for *style*, $\alpha = 0.025$; spontaneous: $F(15, 240) = 3.2$, $p < .0001$, read: $F(15, 240) = 2.5$, $p = .002$). For the spontaneous sentences, post hoc comparisons (Tukey HSD) revealed that 9 out of 120 pairwise speaker comparisons were significant (8%). For the read sentences, 4 out of 120 pairwise comparisons were significant (3%). To test for the simple effect of *style* we processed *t*-tests for every speaker. There were no significant differences in %V between the two styles for all speakers (Bonferroni adjusted for *speaker*, $\alpha = 0.003$).

As for %VO, the model with *speaker* as a random effect provided an improved goodness of fit ($AIC = 3518$, $p = .015$), which means that between-speaker variation was significant. There was a significant effect of *style* with read speech demonstrating an overall higher %VO ($M = 64.6$, $SD = 8$) than spontaneous speech ($M = 63.1$, $SD = 9$) ($AIC = 3518$, $p = .0006$). Results further revealed an interaction of *speaker* $\times$ *style* ($AIC = 3517$, $p = .0003$). To test for the simple effect of *speaker* we processed two ANOVAs: one on the spontaneous sentences and one on the read sentences. Both ANOVAs showed significant effects of *speaker* (Bonferroni adjusted for *style*, $\alpha = 0.025$; spontaneous: $F(15, 240) = 2.3$, $p = .004$, read: $F(15, 240) = 2.8$, $p = .0006$). Post hoc comparisons (Tukey HSD) for each of the ANOVAs revealed that 3 out of 120 pairwise speaker comparisons were significant (2%) for the spontaneous material, and 3 out of 120 comparisons for the read material. To test for the simple effect of *style*, we processed *t*-tests for every speaker (Bonferroni adjusted for *speaker*, $\alpha = .003$). None of the tests showed significant effects.

3.4. How robust are suprasegmental temporal features to channel variability?

We report findings of 16 sentences per speaker that were transmitted over the mobile phone line (the spontaneously produced sentences taken from each speaker's own interview, therefore differing between speakers). As for %VO, the model with *speaker* as random effect performed better than its reduced model ($AIC = 3422$, $p = .0009$), i.e. between-speaker variation was significant. We found a significant difference in %VO across the two channels ($AIC = 3422$, $p < .0001$): Speech transmitted over the mobile phone overall exhibits higher %VO values ($M = 66$, $SD = 8$)

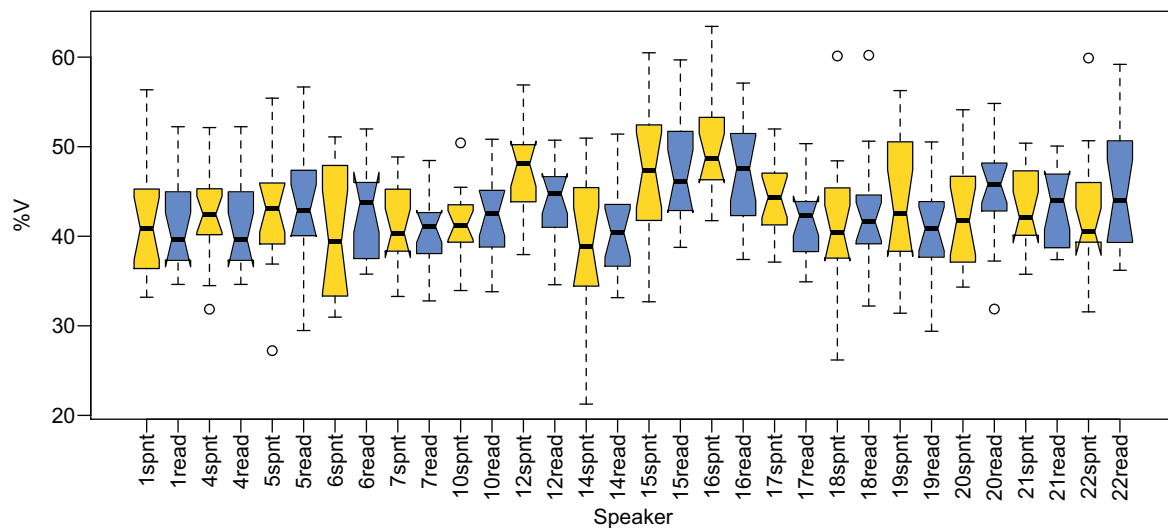


Fig. 3. Boxplots of the 16 speakers' %V (percentage over which speech is vocalic): spontaneous ('spnt', yellow) and read ('read', blue).

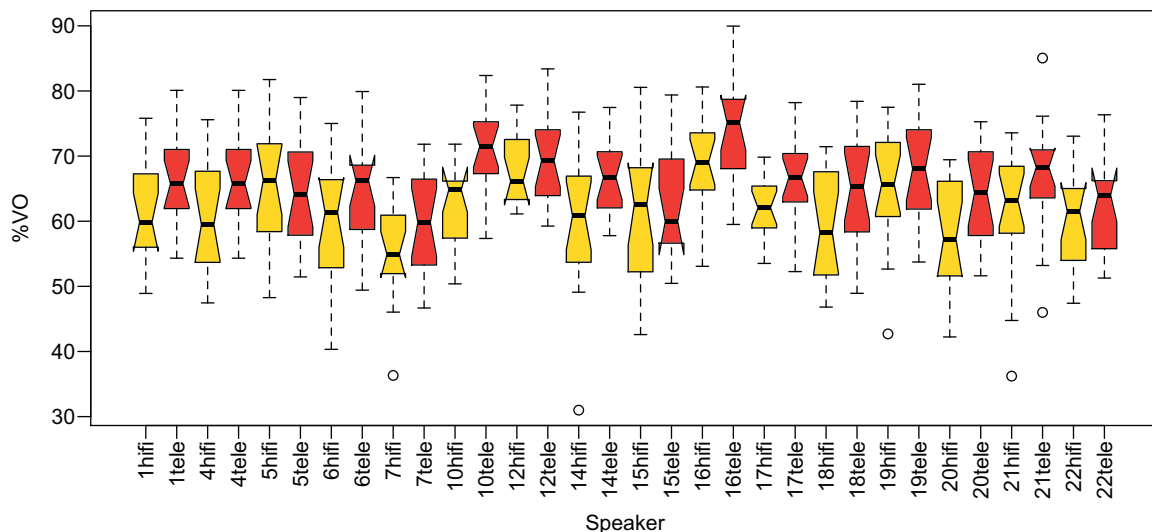


Fig. 4. Boxplots of the 16 speakers' %VO (percentage over which speech is voiced): hifi-recorded sentences ('hifi', yellow) and mobile phone-transmitted sentences ('tele', red).

than speech recorded in a sound-treated booth ($M = 62$, $SD = 9$). There was a significant interaction of *speaker* $\times$ *channel* ($AIC = 3417$, $p < .0001$). Fig. 4 shows the boxplots of the 16 speakers' %VO for the hifi-recorded sentences ('hifi', yellow) and the same sentences transmitted over the mobile phone line ('tele', red).

To test for the simple effect of *speaker* we processed two ANOVAs: one on the hifi-recorded sentences and one on the mobile phone-transmitted sentences. Both ANOVAs showed significant effects of *speaker* (Bonferroni adjusted for *channel*, $\alpha = 0.025$; hifi: $F(15, 240) = 2.8$, $p = .0004$, mobile phone: $F(15, 240) = 3$, $p = .0001$). For the hifi-recorded sentences, pairwise comparisons (Tukey HSD) revealed that 3 out of the 120 comparisons (3%) were significant. For the mobile phone-transmitted sentences, 6 out of the 120 pairwise comparisons (5%) were significant. To test for the simple effect of *channel*, we processed *t*-tests for every speaker (Bonferroni adjusted for *speaker*, $\alpha = 0.003$). We found a significant difference between the two channels only for speaker 10 (hifi: $M = 63$, $SD = 6$; mobile phone: $M = 71$, $SD = 7$; $t(30) = 3.7$, $p = .0009$).

As for VarcoPeak, we obtained no effect of *speaker* ($AIC = -904$, $p = .37$), no significant difference between the two channels ($AIC = -904$, $p = .56$), and no interaction of *speaker* $\times$ *channel* ($AIC = -897$, $p = .11$). Fig. 5 shows the boxplots of the speakers' VarcoPeak in the hifi-recorded sentences ('hifi', yellow) and in the mobile phone-transmitted sentences ('tele', red).

Descriptively, some speakers showed a greater range in mobile phone-transmitted sentences for VarcoPeak (e.g. speaker 1), others showed a smaller range (e.g. speaker 15). Fig. 5 further reveals that a number of speakers demonstrated higher VarcoPeak values in mobile phone-transmitted sentences (e.g. speaker 4) while others showed an opposite trend (e.g. speaker 5).

4. Discussion

In the present study we reported evidence that speakers vary in suprasegmental temporal features. We showed that a selection of these features remained stable in speaking style variability (%VO and %V) and channel variability (%VO).

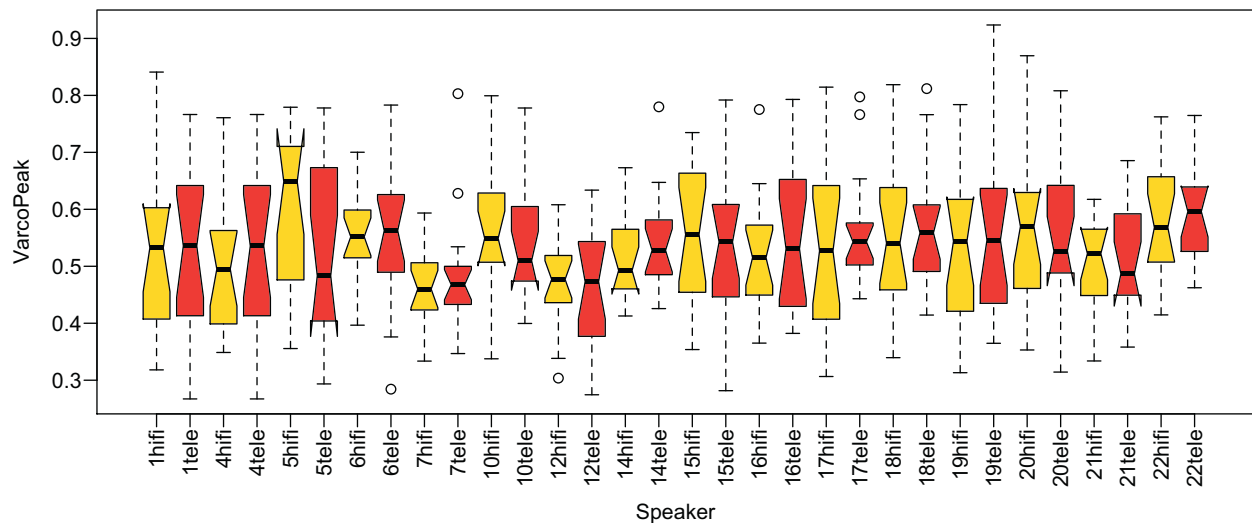


Fig. 5. Boxplots of the 16 speakers' VarcoPeak (rate-normalized standard deviation of syllable-peak-to-syllable-peak interval durations): hifi-recorded sentences ('hifi', yellow) and mobile phone-transmitted sentences ('tele', red).

4.1. Are there between-speaker differences in suprasegmental temporal features?

Results revealed that between-speaker variability is particularly evident in read speech: variability is extensive on the level of vocalic and consonantal, voiced and voiceless, as well as syllable-peak-to-syllable-peak intervals. In spontaneous speech we found less between-speaker variation; from all the measures under observation only %V and %VO showed significant effects of *speaker*. We thus provided strong evidence for speaker-individual suprasegmental temporal characteristics based on a large speaker group that was carefully controlled for accent and dialect variability. We take this to mean that the obtained variability cannot be explained by other factors than speaker-idiolectal or specific articulatory behavior itself.

4.2. Which suprasegmental temporal measure explains most variation between the speakers?

The nominal logistic regressions run on the data showed that %V and %VO reveal the greatest effects of *speaker* (cf. Section 3.2). The radar charts depicted in Figs. 1 and 2 suggest the following scaling of temporal measures: ratio measures such as %V and %VO reveal greater effects than variability measures such as PVI, Varcos, or peak-to-peak. This result is in line with previous research that points to particularly high speaker-individuality in %V [20,23,49] and %VO [49].

It is difficult to explain why exactly %V and %VO show the greatest between-speaker effects. We found that the two measures are moderately correlated. This, as mentioned earlier, can be explained by the fact that vocalic segments are normally voiced and non-vocalic segments can be voiced or voiceless. Yet, current research does not have an adequate understanding as to the determinants that cause %V or %VO to vary. On a conceptual level, we can assume that variability measures are more likely to reflect rhythmicity in speech, while ratio measures portray global vocalic or voicing characteristics. %V may be influenced by other factors such as a speaker-specific way to lengthen final parts of a phrase (phrase-final lengthening [62]). %VO may be a result of voice source characteristics, where creaky or breathy sections in the speech stream are more probable to be picked up as voiceless portions than intervals of sonorant, modal voice portions. Further

work is required to explore the determinants that cause these temporal measures to vary.

The nominal logistic regression model for read speech explained only 8% of suprasegmental temporal variability on the basis of the applied measures. Despite the fact that this low number can be expected given the large size of the corpus (4096 sentences) and given that high R^2 values are rare in categorical data analysis [61], there is room for improvement. It seems conceivable that more refined temporal variability measures will augment differences between speakers. Temporal analyses on a micro level have already revealed distinct speaker-idiosyncratic information: Laan [19] showed speaker-specific durational characteristics of consonants; Amino and Arai [17] report significant between-speaker variability in the temporal energy distribution of nasals. Further studies that specifically examine micro durational variation will need to be undertaken. The manual annotation of our corpus on the segmental and syllable level will allow for highly detailed analyses of additional temporal features. We can examine specific segment combinations, such as nasal/vowel sequences, for example. Or we can investigate the temporal structuring of syllables with different syllabic make-up, for instance. These analyses will likely yield further between-speaker variability in the time domain, which will help maximize discrimination between speakers.

4.3. How robust are suprasegmental temporal features to speaking style variability?

Results in Section 3.3 revealed that speakers differ from one another in %V and %VO in spontaneously produced sentences as well as in their read counterparts. Simple effect tests further showed that these two measures were robust to speaking style variability for every speaker. These two measures thus demonstrate high between-speaker and low within-speaker variability, which is desired for FVC [1]. The generalizability of these results is subject to certain limitations, however. The conversational setting of the spontaneous recordings as well as the sentence material's content is obviously not of a forensic nature. Our study was carried out under laboratory conditions using speech from interviews about everyday topics. We suspect, however, that for our particular experimental framework the application of more crime-oriented contents would not have made much difference.

4.4. How robust are suprasegmental temporal features to channel variability?

Even though it may have seemed trivial to test for channel variability to begin with, the obtained descriptive variability shown in Fig. 4 does justice enough: overall, mobile phone-transmitted speech revealed 4 percentage points higher values for %VO than hifi-recorded speech. Given the interaction of *channel* and *speaker*, however, simple effects were calculated which revealed that these differences are significant for speaker 10 only. Findings on peak-to-peak variability, as measured by VarcoPeak, showed no significant effect of *speaker*, yet also no significant effect of *channel*: syllabic peaks do not discriminate between speakers, but they remain largely unaffected by channel variability. The lack of this effect of *channel* in %VO and VarcoPeak is taken as first evidence that distortions of the type introduced by mobile telephony have little effect on suprasegmental temporal features in general but that they can be significant for particular speakers.

4.5. What are the implications of these results for FVC?

Given the reported evidence for speaker-individuality in suprasegmental temporal features, variables like %V or %VO may find application in FVC either in manual (particularly %V) or automatic (particularly %VO) procedures. Even though some speakers overlap in %VO (see Fig. 4, yellow boxplots), the main effect of *speaker* is still highly significant. This significance must be due to the speakers that vary strongly from each other. This situation is quite common for current FVC parameters such as fundamental or formant frequencies: some speakers vary strongly from each other; others might overlap because of similar larynx and vocal tract shapes. It is possible that in the latter case, however, the same speakers may exhibit differences in suprasegmental temporal features such as %VO or %V. An integration of these additional temporal parameters may thus complement current FVC.

The results reported for speaking style and channel variability also bear implications for FVC. In typical forensic casework, the acoustic trace material is spontaneously produced and the suspect's comparison material is read or spontaneously produced. For this reason it is critical to know if the different speaking styles have an effect on the speech parameter being evaluated in a voice comparison. The knowledge that a speaker's suprasegmental temporal features, in particular %V and %VO, are speaker-specific as well as robust to style variability is vital if a comparison has to be made on material with a mismatch in speaking style. As for channel variability, knowing that the examined suprasegmental temporal features remain relatively stable across channels is critical for FVC since in the majority of cases, forensic trace and suspect material involves telephone-transmitted speech [32]. The points in the signal where voicing starts and ends or where syllable peaks are located seem largely unaffected by channel variability (apart from the case of one speaker). In this regard, we provided evidence showing that suprasegmental temporal features are not strongly influenced by channel variability (see Introduction). This may be an advantage over frequency-based measures.

4.6. The following issues require further research

(1) Even though %V and %VO perform fairly well in explaining suprasegmental temporal variability between speakers, we do not yet have an adequate understanding of what exactly governs variability in these two timing parameters. In the introduction we argued that speaker-specific temporal patterns may be a result of a speaker's anatomical configurations, which in turn are governed by neurological motor patterns in the brain of the speaker. On the other hand, they may also be

caused by the speakers' acquired idiolectal way of articulation. What is the role of speaker anatomy and speaker idiolect in the variability of speech temporal features? Studying twins could help disentangling the underlying reasons for this variability: twin studies allow for a systematic analysis of organic and idiolectal causes for between-speaker variation. With identical twins, we can examine the degree of phonetic differences between the speech of two individuals who are as anatomically similar as possible (i.e. who share 100% of their genes) and whose educational levels and home lives have been relatively the same [63–65]. Because identical twin pairs are matched perfectly for age and sex and as they share the same genes, any differences between them will normally reflect environmental effects. If suprasegmental temporal differences between identical twins could be found, they might be attributed to learned, i.e. idiolectal behavior. Research in this area is relevant because it is not currently known how much of speaker variability in suprasegmental temporal features should be attributed to anatomical and how much to idiolectal factors. Further studies that take speaker anatomy and speaker idiolect into account need to be undertaken. More detailed analyses on these determinants will be critical for an application of suprasegmental temporal measures in forensic casework.

(2) Further studies are needed that address whether between-speaker suprasegmental temporal differences are perceptually salient. Eriksson and Wretling [28] reported preliminary evidence that not even skilled professional impersonators can successfully imitate segmental temporal patterns of a target voice. Most of the currently used variables for FVC, such as f0 or formant frequencies, are highly salient and carry different types of linguistic information. These variables thus vary when speakers disguise their voices [5,66]. Given the presumed lack of salience of suprasegmental temporal features [cf. 28], it seems very likely that such features are difficult to manipulate intentionally.

5. Conclusion

This study set out to examine between-speaker differences in suprasegmental temporal features. The most important findings to emerge from this research are as follows:

1. Speakers exhibited distinct between-speaker variability in all of the examined suprasegmental temporal features.
2. The percentage over which speech is vocalic (%V) and voiced (%VO) revealed the strongest effects of *speaker*.
3. %V and %VO are robust to speaking style variability.
4. %VO is robust to channel variability.

In the present study we provided evidence showing that suprasegmental temporal characteristics can be speaker-specific. Such individuality information may be useful in forensic voice comparison tasks. Insights of the current study are particularly relevant for cases in which there is a mismatch in speaking styles between trace and suspect material, and in cases where the speech signal is degraded by mobile phone transmission.

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